

Telia Advanced connectivity & IoT

- Get the most of your IoT Solutions – Get started and scale on our enabling IoT and AI Platform



IoT

Connectivity
Platform

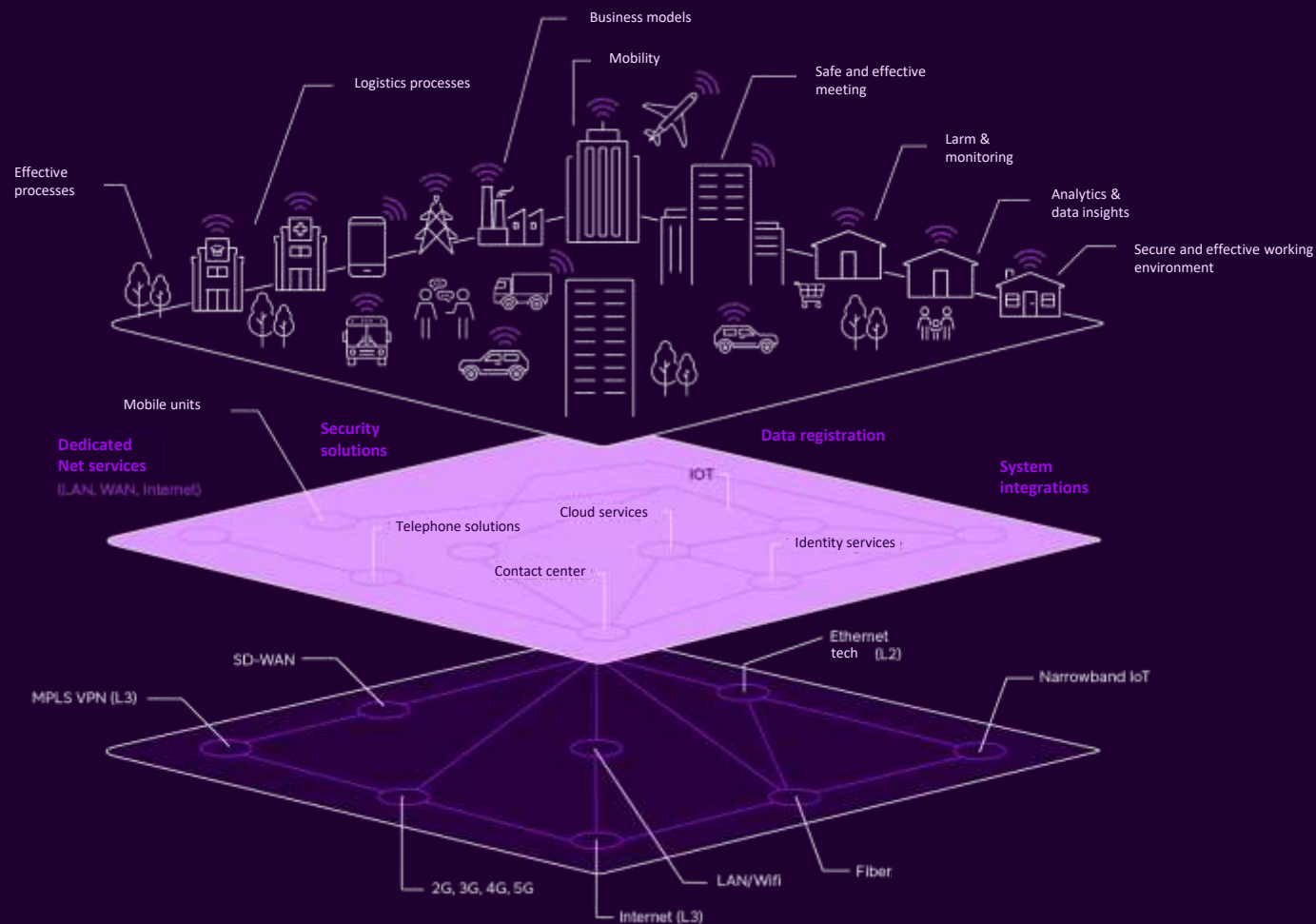
IoT Monitoring &
IoT Platform

Managed IoT
Gateway & Devices

Smart Public
transport &
Logistics

Crowd Insight

Telia enables digitalization



Rikshem & Telia

Rikshem ønsket å skape trygge, komfortable og energieffektive hjem – med full innsikt og kontroll over inneklima og drift.

Men teknologien ble en barriere:

- Hvordan koble tusenvis av sensorer sikkert?
- Hvordan få sanntidsdata som faktisk gir verdi?
- Hvordan skalere fra én bygning til 23 000 leiligheter?

Telia kom inn som teknologipartner og leverte en komplett løsning:

- Sikker tilkobling
- Full enhetsstyring
- Kryptert dataflyt
- Skalerbar dataplattform

Resultatet?

- Proaktiv drift – avvik fanges før de blir problemer
- Store energibesparelser
- Økt leietakertilfredshet
- Raskere og mer presis kundeservice
- Innsikt på tvers av hele porteføljen

Telia hjalp Rikshem med å gå fra ambisjon til realisert verdi.



Danderyds sykehus bruker IoT for å øke energieffektiviteten og forbedre pasientomsorgen

Utfordring:

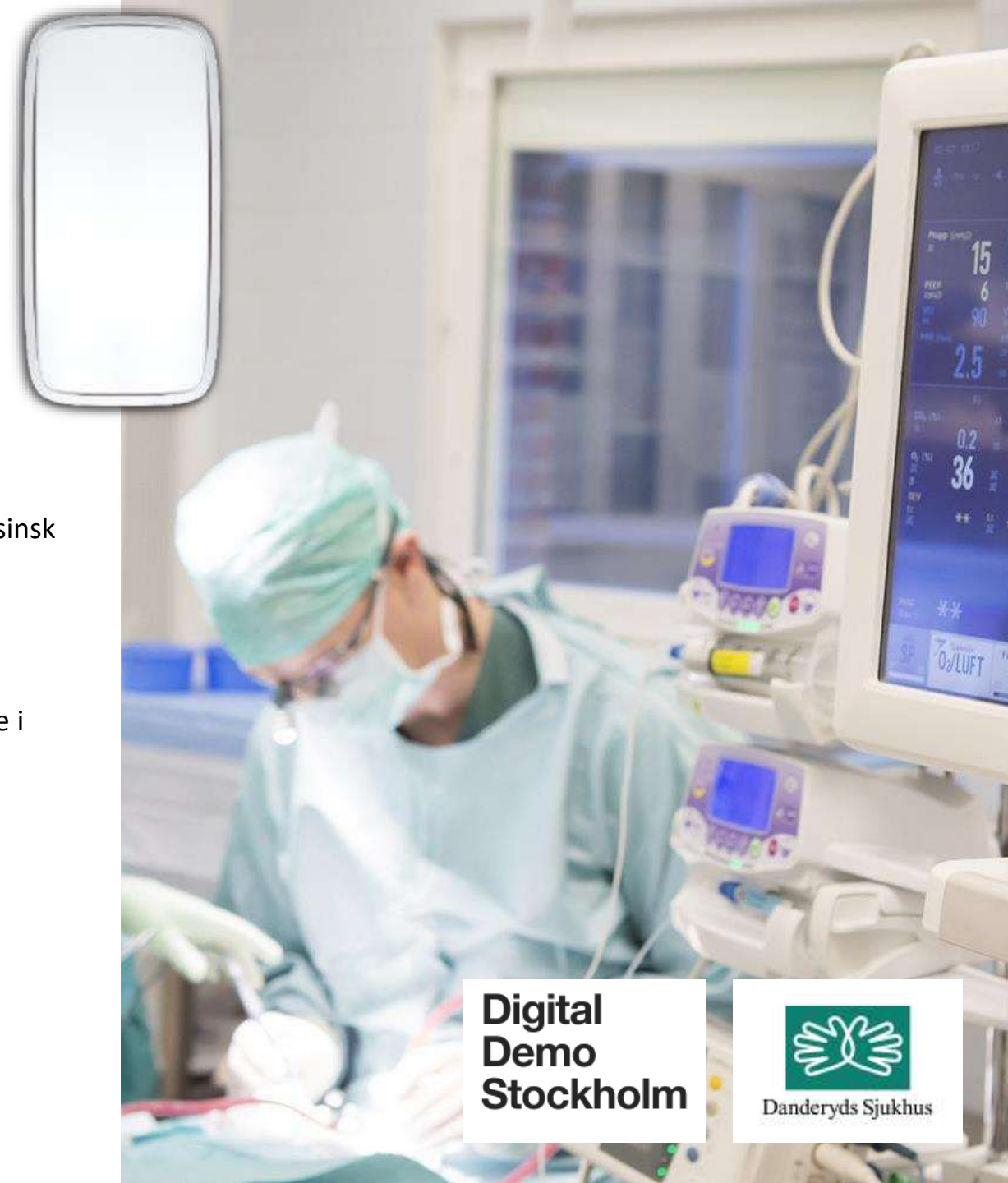
- Sykehus har høyt energiforbruk og strenge krav til inneklima. Feil temperatur eller luftfuktighet kan føre til avlyste operasjoner og at medisinsk utstyr må kasseres eller steriliseres på nytt

Løsning:

- IoT-sensorer ble installert for å måle temperatur, luftfuktighet og lufttrykk. Dataene vises i sanntid på skjermer for sykehuspersonell og brukes til analyse i driften

Verdi:

- Mer effektiv energibruk og lavere CO₂-utslipp
- Tidlig identifisering av feil og stabilt inneklima
- Færre avlyste operasjoner og mindre svinn av utstyr



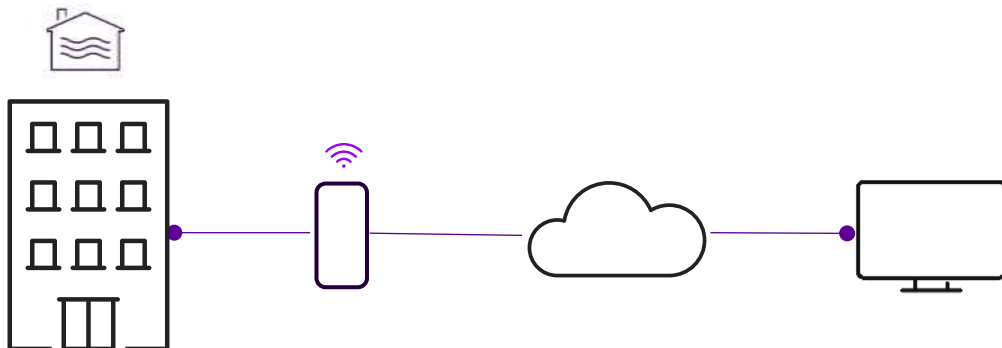
Digital
Demo
Stockholm



Indoor Air Quality Monitoring

Sørg for et sunt inneklima for barn, beboere og ansatte

- Overvåk luftkvaliteten innendørs for å sikre et trygt og helsefremmende miljø i barnehager og pleiehjem.
- Inneklimaindeks (AQI) basert på viktige måleparametere:
 - Temperatur – for komfort og trivsel
 - Luftfuktighet – for å unngå tørr luft og mugg
 - CO₂-ekvivalenter – for å sikre god ventilasjon
 - B-VOC (flyktige organiske forbindelser) – for å redusere skadelige stoffer
 - Lufttrykk – for å forstå luftens bevegelser og påvirkning





Smarter energy management saves consumption and increases tenant comfort

Challenge:

- Stockholm Exergi is a district heating company in Stockholm, heating 90% of Stockholm and 800,000 households

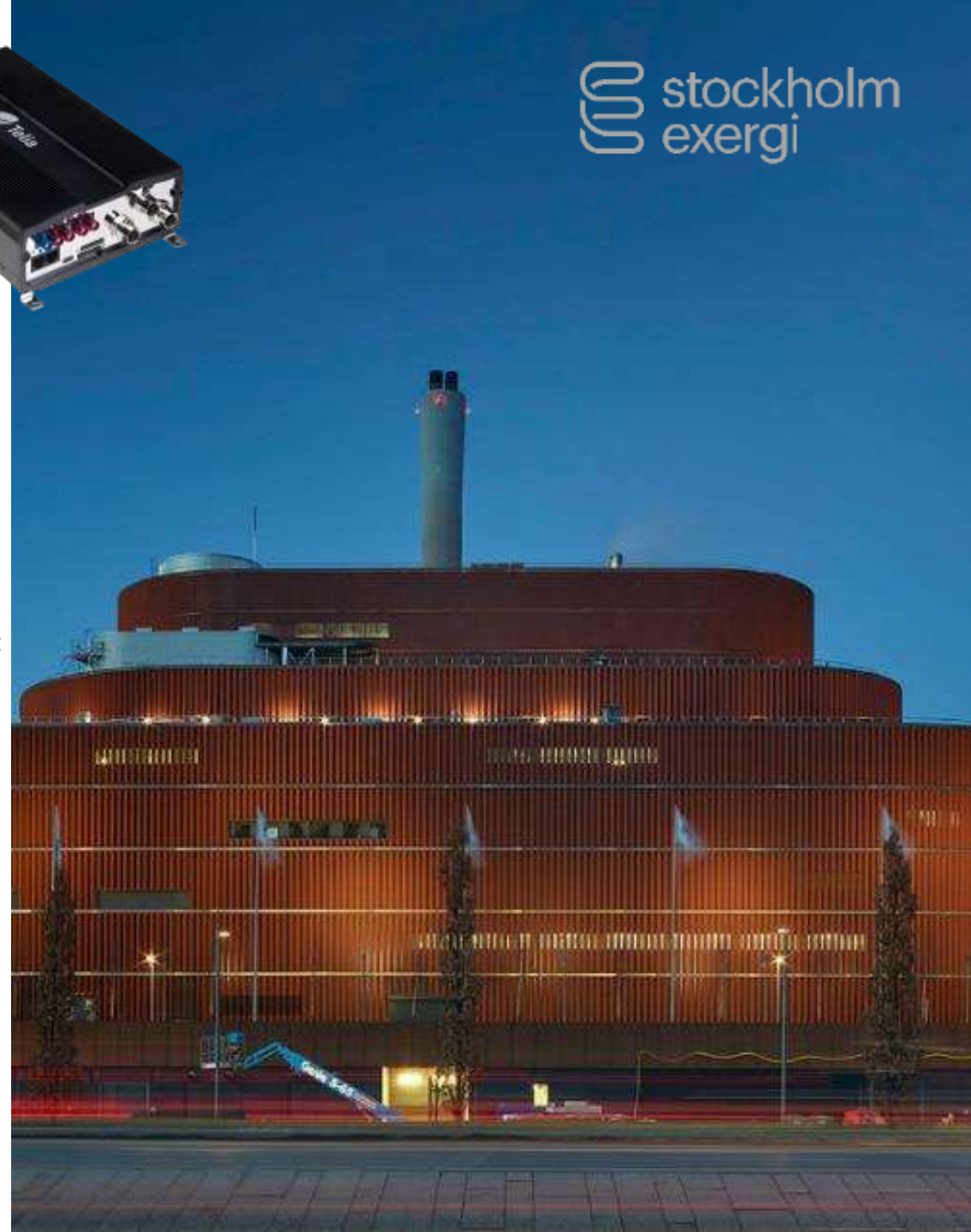
Solution:

- Using modern technology and digital solutions to streamline power and efficiency: connecting 12,000 building with Telia IoT Edge + Connectivity for insights of energy consumption

Value:

- Real-time data from the buildings combined with smart algorithms to predict needs for energy production to optimize the production
- Optimize the distribution of the resources to the buildings where they are needed the most
- Stockholm Exergi is expecting to enable 10% reduced energy consumption
- Adding to this, real-time control, better indoor climate for the tenants and cost savings

[VIDEO: See customer case](#)



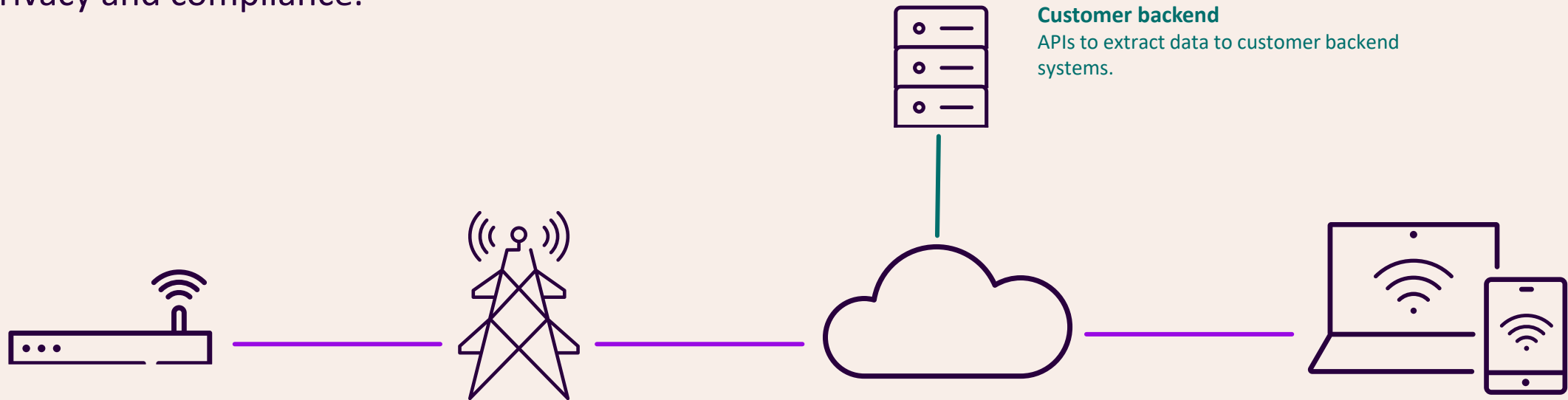


IoT Monitoring & Insights on Telio IoT Platform

Get the most of your IoT Solutions – get started
and scale on our enabling IoT and AI Platform

How it works

Built on Telia's standardized, interoperable modules, ensuring consistent high performance, security, privacy and compliance.



Devices with sensors.

Telia supplies wireless devices (with and without sensors) with pre-installed SIM cards that communicate with our cloud and device management solution.

Indoor and outdoor capabilities

Network & IoT Connectivity

Data points are collected and transmitted over Telia's mobile network.

Telia IoT Platform

The data is transmitted securely to Telia IoT Platform keeping the data securely separated from the public Internet. Data frequency configured to customer needs.

- Operations
- Device management
- Data management

Customer backend

APIs to extract data to customer backend systems.

Management Portal

Device, asset and data management. Option to add dashboards for visualization.



Kundebehov

Flow / Gjennomstrømning

Vannstand innsjøer

Nivåmåling tanker

Lekkasjedeteksjon

Trykk i tank

Luftkvalitet

Temperatur/fukt

Overvåking av pumper

Mobil tilkobling

Internett-tilgang

Trådløst nettverk

Teknikk



Sensorer



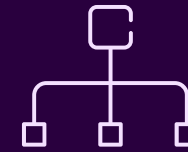
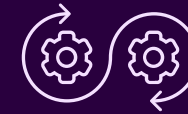
Ruter / Gateway



IoT Edge

Integrasjon & visualisering

Application
Programming
Interface



Scada X

Scada Y

Scada Z



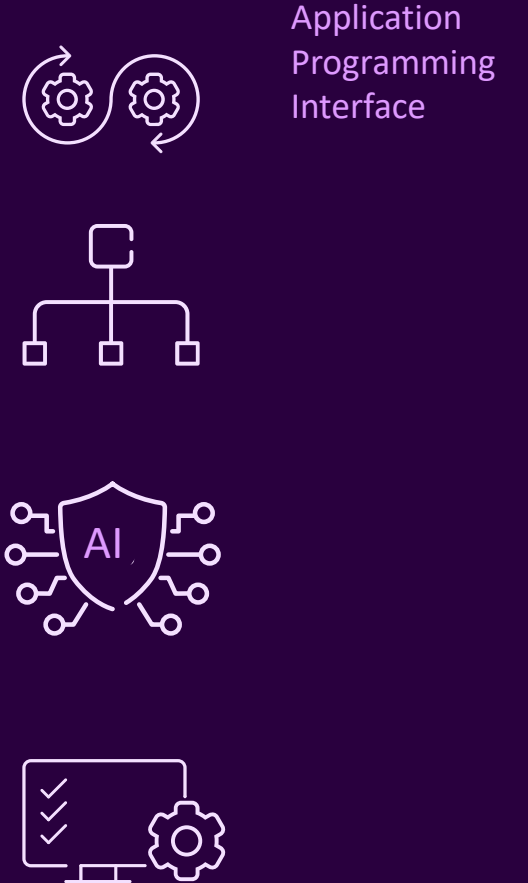
Funksjonelle behov

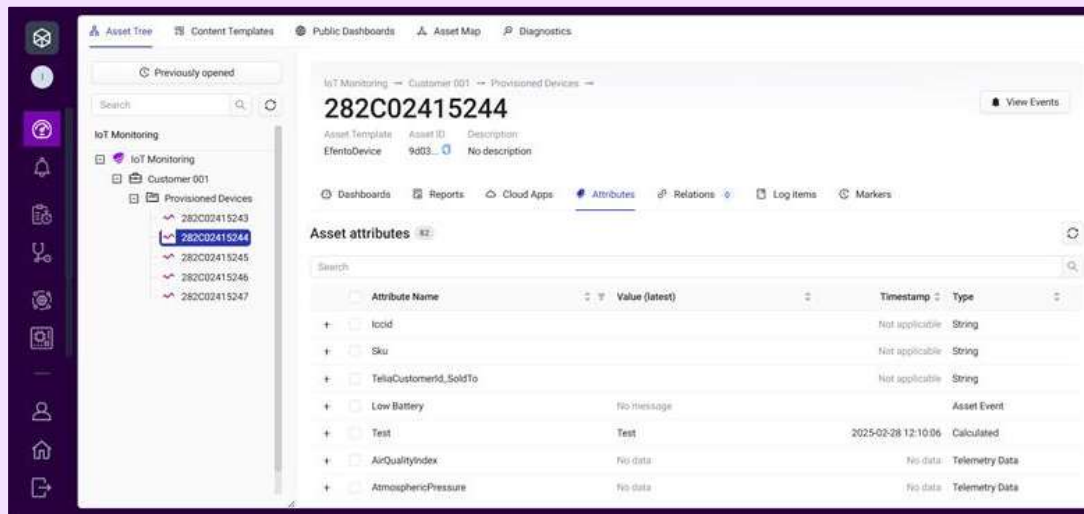
Temperatur	■	
Fuktighet	■	
CO2	■	
4-20mA	■	
Modbus	■	■
Canbus/ODB	■	■
Digital inn/ut	■	■
Analog inn/ut	■	■
Edge prosessering		■
Relé kontroll	■	■
Internett-tilgang	■	■
Trådløst nettverk	■	■
Posisjonering	■	■

Teknikk



Integrasjon & visualisering





Benefits IoT Monitoring



Fast and easy

Bundled solution – device, connectivity and IoT Platform - to get you started fast without a costly, complex and time-consuming onboarding. Simplified installation and activation.



Simplicity at its best

Reducing the complexity in IoT solutions with core IoT solution needs: device, connectivity and data management.



Security & Compliance

Telia adds security from the data point to the cloud, offering REST API's to enable data export secured by authentication and authorization.



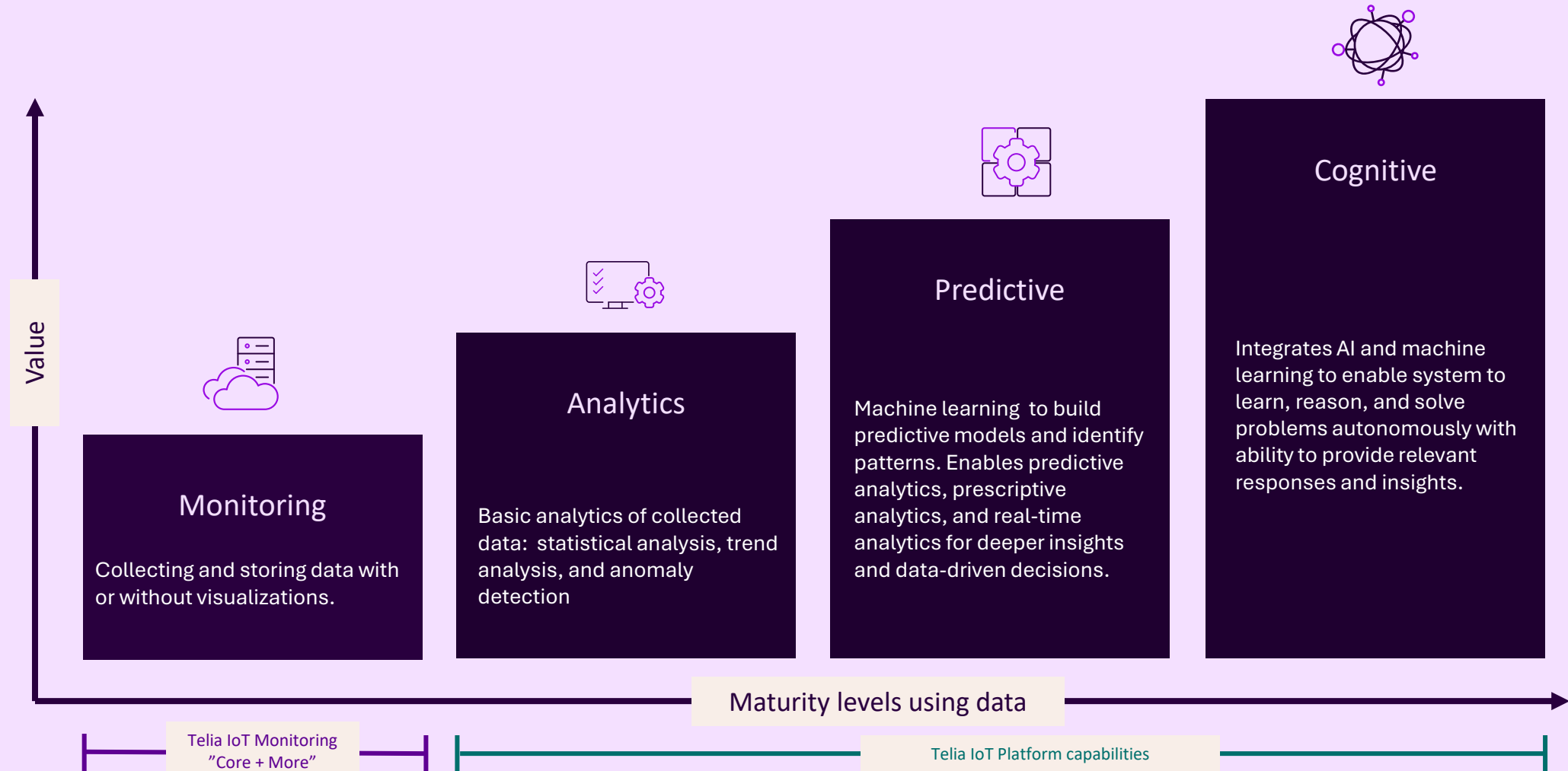
Future proof & scalable

Packaged product on Telia IoT Platform enabling you to easily scale and grow with the wider capabilities.



Where are you on the IoT maturity ladder?

We've got your back every step on your digitalization journey, with the tools that enables you to start and take the next step up the ladder.

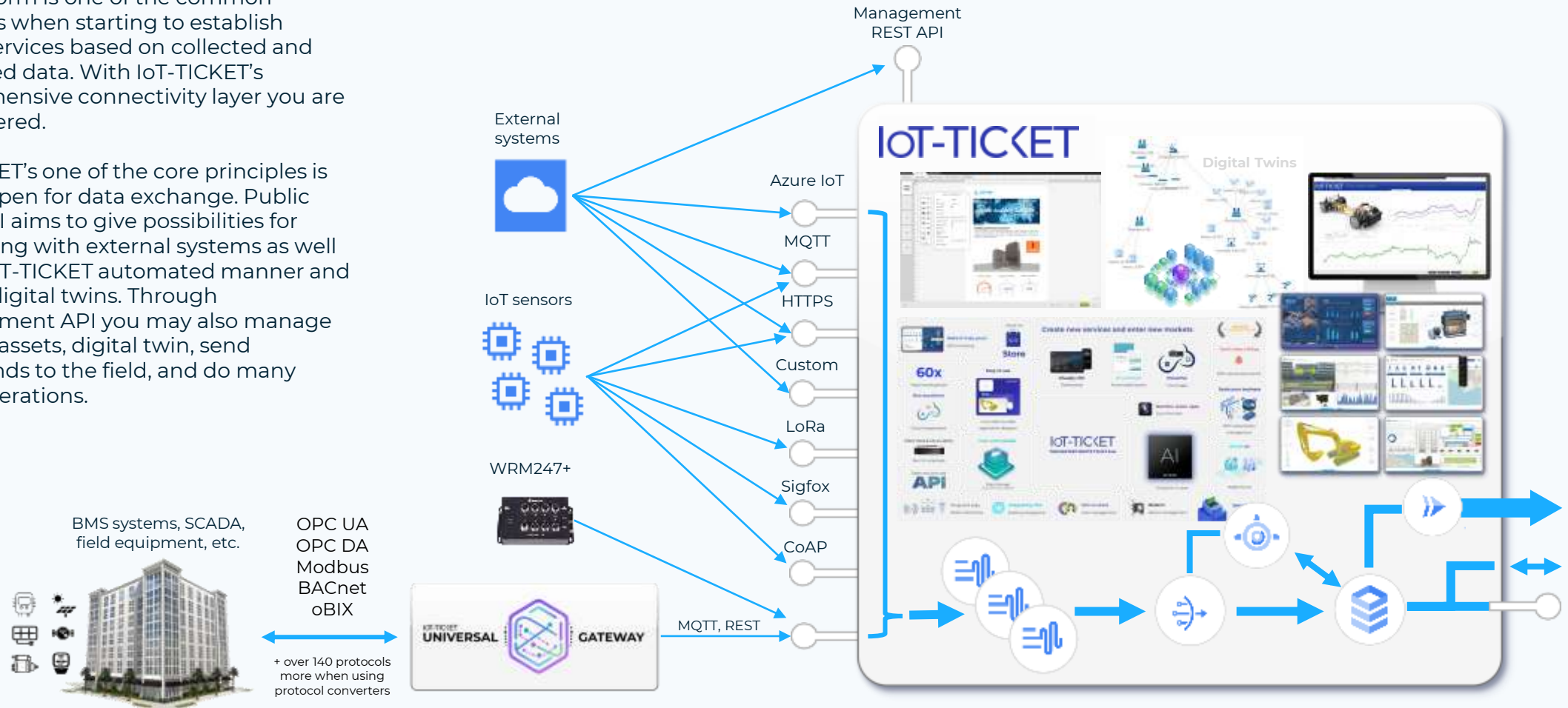


Unlock the full power of IoT with a secure and enabling IoT Platform.



End-to-end connectivity, Data acquisition, and Export with IoT-TICKET platform

- Data exchange between data sources and IoT platform is one of the common concerns when starting to establish digital services based on collected and processed data. With IoT-TICKET's comprehensive connectivity layer you are well covered.
- IoT-TICKET's one of the core principles is to stay open for data exchange. Public open API aims to give possibilities for integrating with external systems as well as use IoT-TICKET automated manner and master digital twins. Through management API you may also manage devices, assets, digital twin, send commands to the field, and do many more operations.



Features:

Optimize data flows and visualization

Acquire data & break down data silos

- Gather valuable data from various sources into one platform.
- Seamlessly connect and utilize IoT devices, sensors, and existing systems.
- Integrate data from other systems with the platform API.

Tools to visualize data in dashboards

- Visualize data with over 120 widgets and data-flow blocks.
- Set automation rules and receive alerts.
- Seamless monitoring and management.
- Maximize efficiency and make data-driven decisions



Features:

Leverage on AI & Machine Learning capabilities



Take the next level with AI/ML

With AI and Machine Learning at your disposal you can transform raw information into actionable intelligence that fuels your growth and inspires new business models.

- Let AI & ML capabilities identify patterns, detect anomalies, and predict trends
- Experience a new level of efficiency, accuracy, and innovation as you unlock the full potential of your IoT ecosystem.

